

Diagnostic Manual

for Development Kit 4

MMSP™ 2

User Manual

MDMP2520FDEB411DIUM

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MP2520F

*Diagnostic Manual for Development Board based MMSP 2
Processor*

Available Board - Development Board 4.1.1

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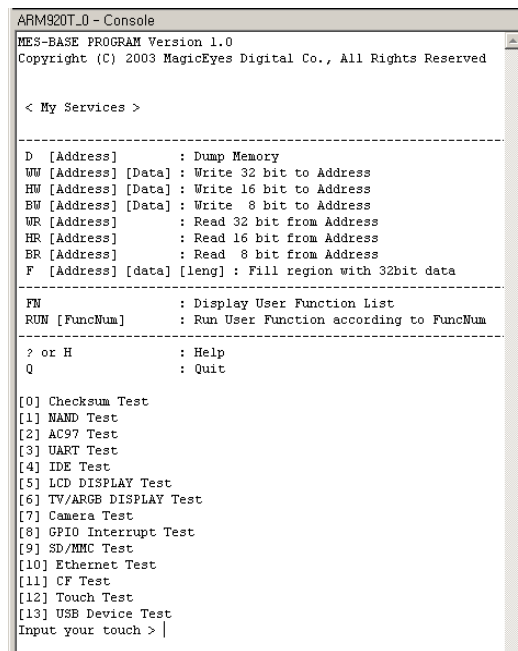
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1. Checksum Test

- Checksum Test is for testing and verifying the stored data in DRAM or specific memory area.
- Result of Checksum test is checksum code.
- Fig 1 is show the Diagnostic Test list.



```

ARM920T_0 - Console
MES-BASE PROGRAM Version 1.0
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< My Services >

-----
D [Address]      : Dump Memory
WW [Address] [Data] : Write 32 bit to Address
HW [Address] [Data] : Write 16 bit to Address
BW [Address] [Data] : Write  8 bit to Address
WR [Address]      : Read 32 bit from Address
HR [Address]      : Read 16 bit from Address
BR [Address]      : Read  8 bit from Address
F [Address] [data] [leng] : Fill region with 32bit data
-----
FN              : Display User Function List
RUN [FuncNum]   : Run User Function according to FuncNum
-----
? or H         : Help
Q              : Quit

[0] Checksum Test
[1] NAND Test
[2] AC97 Test
[3] UART Test
[4] IDE Test
[5] LCD DISPLAY Test
[6] TV/ARGB DISPLAY Test
[7] Camera Test
[8] GPIO Interrupt Test
[9] SD/MMC Test
[10] Ethernet Test
[11] CF Test
[12] Touch Test
[13] USB Device Test
Input your touch > |
    
```

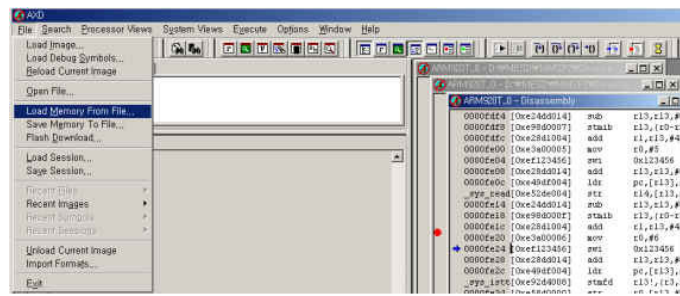
<Fig 1. Diagnostic Test List>

1.1. Command

- [run 0]
- Address : [1000000]
- Size : [1000000]

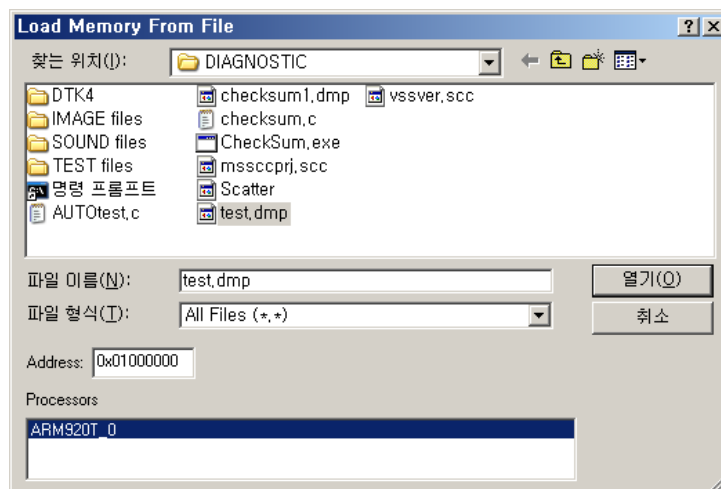
1.2. Checksum Test

- Before execute [run 0], load test file into memory which will be executed. (Refer to Fig 2,3)
(File → LoadMemoryFromFile)



< Fig 2. Load memory from file>

- Press Shift+F5 to **stop** the debugger.
- Select "Load Memory From File" from the "File" menu in the debugger window.
- Open "test.dmp" file in the supplied CD with the kit. (File location is DTK4 → SOFTWARE → PGM → BD_TEST → DIAGNOSTIC)
- Put in address value in the pop up window to define the location of loaded image file. (Example: 0x1000000)
- The status bar in the bottom AXD window shows the memory loading status.
- Wait until loading is completed. (Status bar will be disappeared when loading has completed)
- Press F5 to **start** the debugger.



< Fig 3. Load test file>

- Execute [run 0] command in the console window to invoke the "Checksum Test" mode.
- After execution of "run 0" command, put in the start address of memory to be verified. (example: 0x1000000)
- Press [ENTER] key after put in the size of memory to be tested. (Example: 0x1000000)
- Check Sum will be completed and console window displays the Check Code. (Refer to Fig 4)
- This example shows the Checksum test procedures. Memory check sum is from the memory address 0x01000000 to 0x02000000. Check Sum result is 0x2d573d93

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- You can do it at DOS.
- Execute [checksum test.dmp] command in the DOS.
- Check Sum will be completed and display the check code result. (Refer to Fig 5)

```
Input your touch > run 0
Start [Checksum Test]

ADDRESS(HEX)> 1000000
SIZE(HEX)> 1000000
ADDRESS : 0x01000000 SIZE : 0x01000000
CHECK CODE : 2d573d93

End [Checksum Test]
```

<Fig 4. Checksum Test at Debug Window>

```
E:\WMESD\WMSP2\DTK4\SOFTWARE\PGM_DTK4\BD_TEST\DIAGNOSTIC>checksum test.dmp
file length : 16777216
file length : 01000000
checksum code : 2d573d93
```

<Fig 5. Checksum Test at DOS>

2. NAND Test

- NAND Test program is for testing the NAND Flash read/write operation.
- To access NAND Flash memory, it requires the initializing the NAND memory firstly.

2.1. Command

- [run 1]
- [mt]
- [ni]
- [pr 0 0 1000000 4000]
- [rd 0 0 1000000 4000]

2.2. NAND Flash Test

- Execute [run 1] command in the console window to invoke the “NAND Test” mode.

2.2.1. NAND Flash Write Mode (Store Data into NAND Flash memory)

[MT → NI → PR]

- Execute [MT] command after [run 1] command. [MT] command is for Make NBAT information and Initialization.
- If NAND Flash memory initialization completed successfully, then you will see Fig 6 in console window.
 - “bank” means the bank address of NAND Flash memory (Default bank value is “0”)
 - “maker code” means the Flash Memory manufacturer name. (Default vendor name is Samsung)
 - “device code” means the Flash Memory part number.
 - “description” shows the configuration of NAND Flash memory (example shows 8 bit data bus width and 64MB memory size)
- Execute [NI] command after [MT]. (Users have to execute [NI] command always before the [PR] or [RD] command.)
- When [NI] command has completed successfully, console window displays “INIT OK”
- After [NI] command completed successfully, execute the [PR] command to start the write process.
- To execute the “PR” command, users have to put in 4 arguments as in below. (Refer to Fig6)
- Example : [PR arg1 arg2 arg3 arg4]

- **arg1** : bank information
- **arg2** : Start address of NAND flash memory to be written.
- **arg3** : Start address of main memory (SDRAM) to be copied to NAND Flash memory
- **arg4** : Data size to be copied
- When the process reaches 100%, it means the write process from SDRAM into NAND Flash memory has completed.
- If users want to execute [PR] command again, [NI] command must be executed once again before the next [PR] command.

2.2.2. NAND Flash Read Mode

- NAND Flash Read mode procedure is as follows.
[MT → NI → RD]
- [RD] command has 4 arguments as in below.
- Example : [RD arg1 arg2 arg3 arg4]
 - **arg1** : bank information
 - **arg2** : Start address of NAND flash memory to be copied.
 - **arg3** : Start address of main memory (SDRAM) to be written from NAND Flash memory
 - **arg4** : Data size to be copied
- Example shows the bank number is "0", NAND Flash start address is "0x8000", memory address is "0x2100000" and data size is "0x100000".
- When users define the "arg2" and "arg4", the address offset must be 0x4000.
- If address value is not divided by 0x4000, error message will be displayed.
- For the "arg3", it also requires the address by word (4 byte) basis.
- Refer to Fig 7
- When process reaches 100%, the read process from NAND Flash memory has completed.
- If users want to execute [RD] command again, [NI] command must be executed once again before the next [RD] command.

```

< NAND Test >

-----
MT : Make NEAT information & Initialization.
NI : Pre-command before Main-command
-----

PR [Bank] [NFAddr] [MemAddr] [ByteSize] : Write mode of NAND Flash
RD [Bank] [NFAddr] [MemAddr] [ByteSize] : Read mode of NAND Flash
[NFAddr] : address of NAND Flash. 0x4000 Byte Align
[MemAddr] : address of memory. 4Byte Align
[ByteSize] : size of data. 0x4000 Byte Align
-----

? or H : Help
Q : Quit

NAND > mt
bank : 0
maker code : EC --> SAMSUNG
device code : 76 --> K9K1208U0A
description : 8bit, 64MB size NAND Flash
--> OK

NAND > ni
K9K1208U0A INIT OK

NAND > pr 0 0 2100000 40000
NAND Write : [50.0] Percent finished.
NAND Write : [100.0] Percent finished.
OK
NAND >

```

<Fig 6. NAND Flash Write Test>

```

< NAND Test >

-----
MT : Make NEAT information & Initialization.
NI : Pre-command before Main-command
-----

PR [Bank] [NFAddr] [MemAddr] [ByteSize] : Write mode of NAND Flash
RD [Bank] [NFAddr] [MemAddr] [ByteSize] : Read mode of NAND Flash
[NFAddr] : address of NAND Flash. 0x4000 Byte Align
[MemAddr] : address of memory. 4Byte Align
[ByteSize] : size of data. 0x4000 Byte Align
-----

? or H : Help
Q : Quit

NAND > mt
bank : 0
maker code : EC --> SAMSUNG
device code : 76 --> K9K1208U0A
description : 8bit, 64MB size NAND Flash
--> OK

NAND > ni
K9K1208U0A INIT OK

NAND > rd 0 8000 2100000 100000
NAND Read : [12.5] Percent finished.
NAND Read : [25.0] Percent finished.
NAND Read : [37.5] Percent finished.
NAND Read : [50.0] Percent finished.
NAND Read : [62.5] Percent finished.
NAND Read : [75.0] Percent finished.
NAND Read : [87.5] Percent finished.
NAND Read : [100.0] Percent finished.
OK

```

<Fig 7. NAND Flash Read Test>

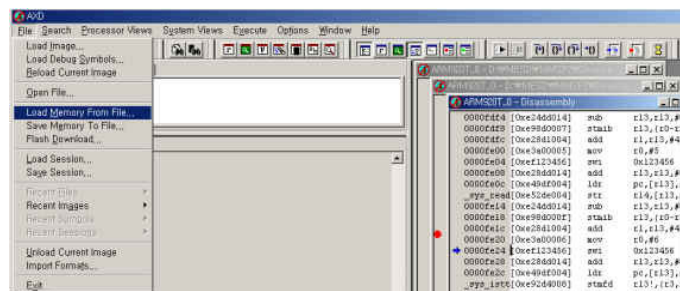
3. AC97 Test

3.1. Command

- [run 3]
- After project stop, Load to PCM file.
- [ac97 1 2 1000000 338d08]
- You can hear to Audio output.

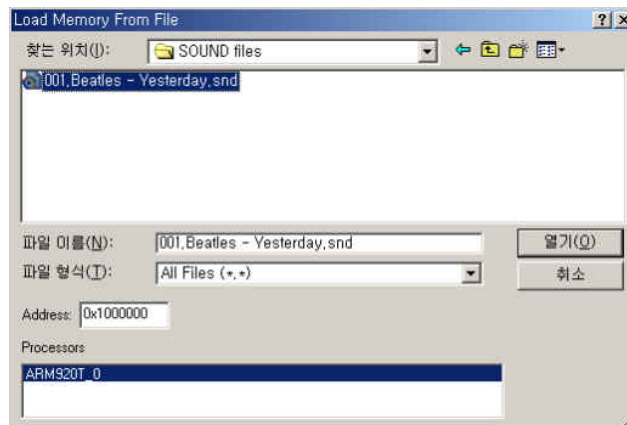
3.2. AC97 Test

- Execute [run 4] command to start AC97 test mode.
- Load sound file into memory which will be executed. (Refer to Fig 8, Fig 9)
(File → LoadMemoryFromFile)



<Fig 8. Load memory from file>

- h. Press Shift+F5 to stop the debugger.
- i. Select “Load Memory From File” from the “File” menu in the debugger window.
- j. Open “001.Beatles - Yesterday.snd” file in the supplied CD with the kit. (File location is DTK4 → SOFTWARE → PGM → BD_TEST → DIAGNOSTIC → SOUND FILES.)
- k. Put in address value in the pop up window to define the location of loaded image file.
(Example: 0x1000000)
- l. The status bar in the bottom AXD window shows the memory loading status.
- m. Wait until loading is completed. (Status bar will be disappeared when loading has completed)



<Fig 9. Load Audio File>

- Execute “AC97” command. (The “AC97” command has 4 arguments as in below.)

[AC97 arg1 arg2 arg3 arg4]

- **arg1[mode]** : “0” is for Line in test. “1” is for Wave out test.
 - **arg2[sampling]** : “0” is for 44.1KHz, “1” is for 32KHz, “2” is for 22.05KHz.
 - **arg3[memory address]** : Start address of stored sound file.
 - **arg4[filesize]** : Data size to be played (Unit is Hexadecimal).
- Example shows 1) File size is 0x338d08, 2) Memory address is 0x1000000 and 3) Wave Out test mode. When successfully completed, Yesterdays will be played.

```

Input your touch > run 2
Start [AC97 Test]

This Function tests the AC97

< AC97 Test >
-----
AC97 [Mode] [Sampling] [MemAddr] [FileSize]: Run AC97
-----
[Mode]      : 0 -> LineIN test, 1 -> WaveOUT test
[Sampling]  : sampling rate. 0->44.1kHz, 1->32.0kHz, 2->22.05kHz
[MemAddr]   : address of memory. 4Byte Align
[FileSize]  : size of wave file.
-----
all options are hexadecimal value
? or H      : Help
Q           : Quit

AC97 > ac97 1 2 1000000 338d08
AC97 test end
OK
AC97 >

```

<Fig 10. AC97 Test>

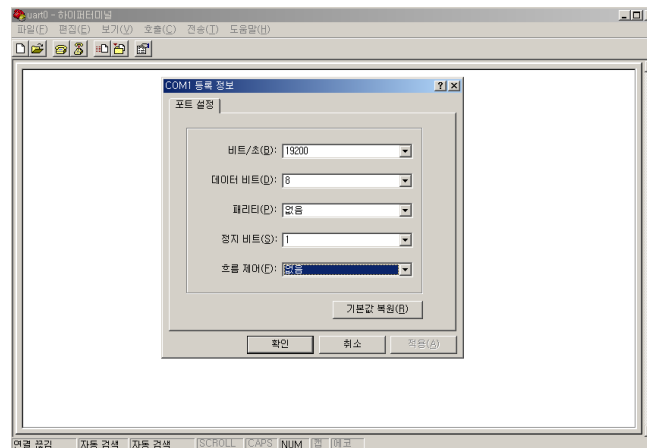
4. UART Test

4.1. Command

- [run 3]
- [UART 0 1]
- Press Any key.
- To stop the UART test, press the “Shift+Q” in the hyper-terminal. [Q]

4.2. UART Test

- Execute [run 3] to start UART test.
- The [UART] command has 2 arguments as in below.
[UART arg1 arg2]
 - **arg1(Channel)** : Number of UART channel.
 - **arg2(Speed)** : Transmit Speed. 0 : 9600bps, 1: 19200bps, 2 : 38400 bps
- Example shows UART Channel 0 with transmit speed 19,200 bps mode.
- Simplest test method is using the hyper terminal with a Personal Computer.
- To use the hyper-terminal, transmit bit rate between the board and PC must be same and other settings are as follows.
- Data Bit is “8” bit, Parity is “no” parity, Stop bit is “1” and flow control is “no”.
- Data put into the Hyper-terminal will be transferred to the development kit thru a UART interface and UART test program will display the received data on the console window as RxData[x].
- Meanwhile, the console window will re-send the received data to the hyper-terminal and displays TxData[x] on the console window as in below.
- Quit (shift + q)



<Fig 11. Hyper Terminal window>

```

Input your touch > run 3
Start [UART Test]

This Function tests the Uart.

< Uart Test >
-----
UART  [CH] [SPEED]  : Receive and Transmit test of UART
-----
          [CH]      : Channel of UART. support 0 ~ 2
          [SPEED]    : Speed of UART. support 9600bps ~ 115200 bps
                      0 -> 9600   bps, 1 -> 19200   bps, 2 -> 38400   bps
-----
all options are hexadecimal value
? or h                                : Help
q                                      : Quit

uart > uart 0 1
UART CHANNEL 0, 19200 bps
check point 1
RxData[0a]
TxData[0a]
RxData[0b]
TxData[0b]
RxData[0c]
TxData[0c]
RxData[0d]
TxData[0d]
RxData[0e]
TxData[0e]

```

<Fig 12. UART Test>

5. IDE Test

5.1. Command

- [run 4]
- [init 0]
- [write 0 0 0 1 0]
- [read 0 0 0 1 1000000]
- [writes 0 7 80000 30 4 2100000]
- [reads 0 7 80000 30 4 2200000]
- [dwrite 0 7 80000 30 2100000]
- [dread 0 7 80000 30 2200000]
- [dwrites 0 7 80000 30 4 2100000]
- [dreads 0 7 80000 30 4 2200000]

5.2. IDE Test

- Execute [run 4] to start IDE test mode.
- IDE test mode requires Initialization process. Execute "INIT" command.
- The "INIT" command has 1 arguments as in below.

[INIT arg1]

- arg1 [DriveNum] : Driver Number which is connect. "0" or "1" (default "0")
- When initialization completed successfully, external storage information will be displayed on the screen.

5.2.1. PIO Mode Command

- To write data thru the PIO mode, execute "write" command.

[WRITE arg1 arg2 arg3 arg4 arg5]

- arg1 [DriveNum] : Number of Driver to be used. (or being used)
- arg2 [Head] : Head number of the drive
- arg3 [Track] : Track number of the drive.
- arg4 [Sector] : Sector number of the drive.
- arg5 [Addr] : Address of memory to be copied into HDD.
- To read the data thru PIO read mode, execute [read] command.
- The [read] command has 5 arguments and read does by 1 sector as in below.

[READ arg1 arg2 arg3 arg4 arg5]

- arg1 [DriveNum] : Number of Driver to be used. (or being used)
 - arg2 [Head] : Head number of the drive to be read
 - arg3 [Track] : Track number of the drive to be read
 - arg4 [Sector] : Sector Number of the drive to be read
 - arg5 [Addr] : Address of memory to be copied from HDD.
- When “READ” command has executed, read data from HDD will be displayed on the console screen.

```

Input your touch > run 4
Start [IDE Test]

Status Register : 0xe5

< IDE Test >
-----
INIT  [DriveNum]      : IdentifyDrive
READ  [DriveNum] [HEAD] [TRACK] [SECTOR] [ADDR]
WRITE [DriveNum] [HEAD] [TRACK] [SECTOR] [ADDR]
-----
? or H                               : Help
Q                                     : Quit

Status Register : 0xd0
IDE > init 0
Manufacture : HTS424030M9AT00
Heads      : 16
Tracks     : 16383
Sector     : 63
Ok
IDE >

```

<Fig 13. Init Command for IDE Test>

```

IDE > write 0 0 0 1 0
IDE_UTIL_SetPIO:0,0
Ok
IDE > read 0 0 0 1 1000000
IDE_UTIL_SetPIO:0,0
00000000 : 00 00 00 00 18 f0 9f e5 : 18 f0 9f e5 18 f0 9f e5 .....
00000010 : 18 f0 9f e5 00 00 a0 e1 : 18 f0 9f e5 18 f0 9f e5 .....
00000020 : 40 00 00 00 80 00 00 00 : 84 00 00 00 88 00 00 00 0.....
00000030 : 8c 00 00 00 00 00 00 00 : b0 18 01 00 90 00 00 00 .....
00000040 : 4c 00 9f e5 d1 f0 21 e3 : 00 d0 40 e2 d2 f0 21 e3 L.....!
00000050 : 40 df 40 e2 d7 f0 21 e3 : 80 df 40 e2 db f0 21 e3 0.0.....!
00000060 : c0 df 40 e2 d3 f0 21 e3 : 40 de 40 e2 00 00 0f e1 ..0.....!
00000070 : c0 00 80 e3 00 f0 2f e1 : 06 00 00 eb fe ff ff ea ...../.....
00000080 : fe ff ff ea 0e f0 a0 e1 : fe ff ff ea fe ff ff ea .....
00000090 : fe ff ff ea 00 00 00 01 : 90 80 8f e2 0f 00 98 e8 .....
000000a0 : 08 00 80 e0 08 10 81 e0 : 08 20 82 e0 08 30 83 e0 .....0..
000000b0 : 01 b0 40 e2 01 c0 42 e2 : 01 00 50 e1 0e 00 00 0a ..0...B...P....
000000c0 : 70 00 b0 e8 05 00 54 e1 : fa ff ff 0a 01 00 14 e3 p.....T.....
000000d0 : 0b 40 84 10 01 00 15 e3 : 0b 50 85 10 02 00 15 e3 .0.....P....
000000e0 : 09 50 85 10 03 50 c5 e3 : 04 60 56 e2 04 70 94 24 .P...P...V...p.$
000000f0 : 04 70 85 24 fb ff ff 8a : ee ff ff ea 03 00 52 e1 .p.$.....R.
00000100 : b6 69 00 0b 00 70 a0 e3 : 30 00 b2 e8 01 00 14 e3 .1...p..0.....
00000110 : 0c 40 84 10 02 00 14 e3 : 09 40 84 10 03 40 c4 e3 .0.....0...0..
00000120 : 04 50 55 e2 04 70 84 24 : fc ff ff 8a f2 ff ff ea .PU..p.$.....
00000130 : 24 ff 01 00 60 ff 01 00 : 60 ff 01 00 70 ff 01 00 $.p.....p...
00000140 : 04 e0 2d e5 00 c0 a0 e3 : 0c e0 a0 e1 09 00 00 ea ..p.....
00000150 : 0e 20 a0 e1 0e 10 a0 e1 : 01 30 a0 e1 01 10 81 e2 .....0.....
00000160 : 64 00 51 e3 fb ff ff ba : 01 20 82 e2 7d 00 52 e3 d.Q.....).R.
00000170 : f7 ff ff ba 01 c0 8c e2 : 00 00 5c e1 f3 ff ff ba .....p.....
00000180 : 04 f0 9d e4 7c 40 2d e9 : 00 60 a0 e1 02 50 a0 e1 ....(0-...P..
00000190 : 01 40 a0 e1 04 20 8d e2 : 00 10 a0 e3 8a 00 a0 e3 .0.....
000001a0 : 36 2f 00 eb 04 00 dd e5 : 1f 20 00 e2 04 00 dd e5 6/.....
000001b0 : a0 12 a0 e1 eb 0f 8f e2 : 5d 61 00 eb 80 20 a0 e3 .....ja.....
000001c0 : ba 10 a0 e3 8a 00 a0 e3 : fc 2e 00 eb 0a 00 a0 e3 .....
000001d0 : da ff ff eb 46 20 a0 e3 : 6c 10 a0 e3 8a 00 a0 e3 ....F...l.....
000001e0 : f6 2e 00 eb 04 08 85 e1 : 00 00 56 e3 95 00 00 0a .....V.....
000001f0 : 01 00 56 e3 f3 00 00 0a : 02 00 56 e3 4f 01 00 1a ..V.....V.O...
Ok
IDE >

```

<Fig 14. Read/Write Command for IDE Test>

- To write data thru the PIO mode, execute “writes” command.

[WRITES arg1 arg2 arg3 arg4 arg5 arg6]

- arg1 [DriveNum] : Number of Driver to be used. (or being used)
 - arg2 [Head] : Head number of the drive
 - arg3 [Track] : Track number of the drive.
 - arg4 [Sector] : Sector number of the drive.
 - arg5 [Sector Cnt] : Sector Count Number.
 - arg6 [Addr] : Address of memory to be copied into HDD.
- To read the data thru PIO read mode, execute [read] command.
 - The [read] command has 6 arguments and read does by sector count number as in below.
- [READS arg1 arg2 arg3 arg4 arg5 arg6]**
- arg1 [DriveNum] : Number of Driver to be used. (or being used)
 - arg2 [Head] : Head number of the drive to be read
 - arg3 [Track] : Track number of the drive to be read
 - arg4 [Sector] : Sector Number of the drive to be read
 - arg5 [Sector Cnt] : Sector Count Number.
 - [Addr] : Address of memory to be copied from HDD.
- When “READ” command has executed, read data from HDD will be displayed on the console screen.

5.2.2. DMA Mode Command

- To write data thru the DMA mode, execute “dwrite” command.
- The [dwrite] command has 5 arguments and read does by 1 sector as in below.

[DWRITE arg1 arg2 arg3 arg4 arg5]

- arg1 [DriveNum] : Number of Driver to be used. (or being used)
 - arg2 [Head] : Head number of the drive
 - arg3 [Track] : Track number of the drive.
 - arg4 [Sector] : Sector number of the drive.
 - arg5 [Addr] : Address of memory to be copied into HDD.
- To read the data thru DMA read mode, execute [read] command.
 - The [read] command has 5 arguments and read does by 1 sector as in below.

[DREAD arg1 arg2 arg3 arg4 arg5]

- arg1 [DriveNum] : Number of Driver to be used. (or being used)
- arg2 [Head] : Head number of the drive to be read
- arg3 [Track] : Track number of the drive to be read

- arg4 [Sector] : Sector Number of the drive to be read
- arg5 [Addr] : Address of memory to be copied from HDD.
- To write data thru the DMA mode, execute “dwrites” command.
[DWRITES arg1 arg2 arg3 arg4 arg5 arg6]
 - arg1 [DriveNum] : Number of Driver to be used. (or being used)
 - arg2 [Head] : Head number of the drive
 - arg3 [Track] : Track number of the drive.
 - arg4 [Sector] : Sector number of the drive.
 - arg5 [Sector Cnt] : Sector Count Number.
 - Arg6 [Addr] : Address of memory to be copied into HDD.
- The read command has 6 arguments and read does by sector count number as in below.
[DREADS arg1 arg2 arg3 arg4 arg5 arg6]
 - arg1 [DriveNum] : Number of Driver to be used. (or being used)
 - arg2 [Head] : Head number of the drive to be read
 - arg3 [Track] : Track number of the drive to be read
 - arg4 [Sector] : Sector Number of the drive to be read
 - arg5 [Sector Cnt] : Sector Count Number.
 - arg6 [Addr] : Address of memory to be copied from HDD.

5.2.3. Verify Test

- This command is comparison between Data of Source Address with Data of Destination Data. If a address has different data, then it is display out Data value and Address
- This Command is execute a verification as Sector Count
- This command has 3 arguments.

[v 2100000 2200000 80]

- arg1 [Source Address] : Source Address is included Data
- arg2 [Dest Address] : Destination Address is empty or it has garbage data
- arg3 [Sector Count] : Sector Count Number of verification

6. LCD Display Test

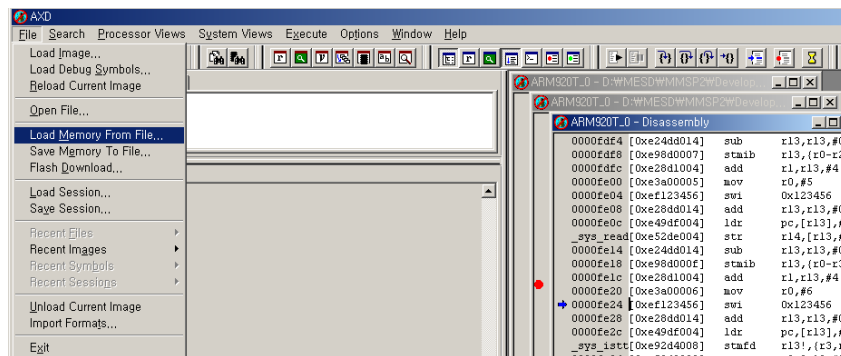
- Display Test draws the stored graphic image in memory onto external LCD.
- Built in LCD module in the development kit has 800 x 480 resolution and data format between MMSP2 to LCD module is RGB666.

6.1. Command

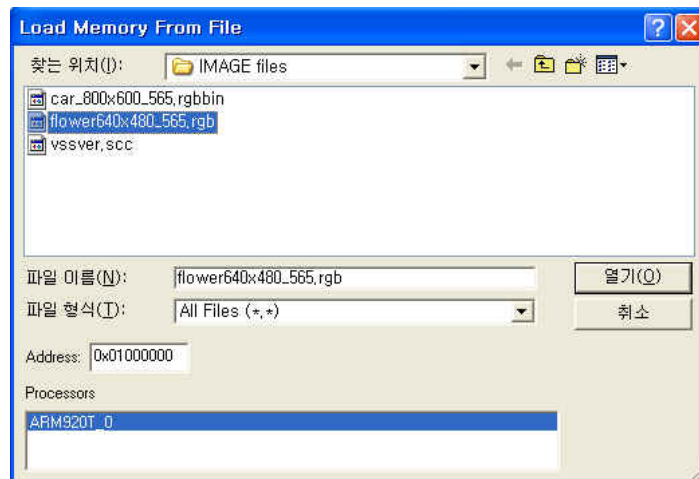
- [run 5]
- Image File load to the memory
- [LCD 1000000]

6.2. LCD Display Test

- Execute [run 5] to start “DISPLAY test” mode.
- Load image file into memory which will be displayed onto LCD module. (Refer to Fig 15)
 - a. Press Shift+F5 to stop the debugger.
 - b. Select “Load Memory From File” from the “File” menu in the debugger window.
 - c. Open “car_800x480.bin” file in the supplied CD with the kit. (File location is DTK4 → SOFTWARE → PGM → BD_TEST → DIAGNOSTIC → IMAGE FILES.)
 - d. Put in address value in the pop up window to define the location of loaded image file. (Example: 0x1000000)
 - e. The status bar in the bottom AXD window shows the memory loading status.
 - f. Wait until loading is completed. (Status bar will be disappeared when loading has completed)

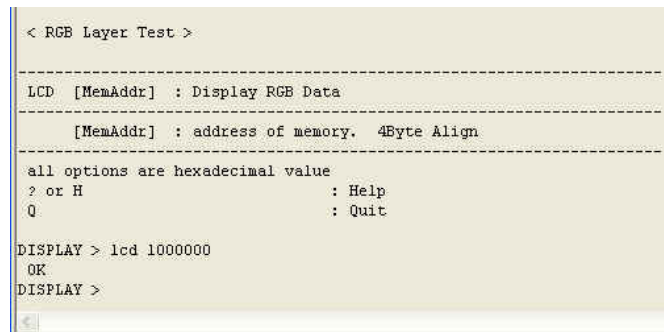


<Fig 15. Load Memory From File for Display Test>



<Fig 16. Load memory from file>

- Execute LCD command for Display test mode.
- [LCD] command has a argument
[LCD arg1]
 - arg1(**MemAddr**) : MemAddr defines memory address of loaded image file or to be displayed image file.
- When DISPLAY test mode completed correctly, LCD panel will show a flower garden.



<Fig 17. LCD Display Test>

7. TV/ Analog RGB/YPbPr Display Test

- Development Board 4.1.1 support to RGB Monitor output and TV(Composite) output, YPbPr output.
- Development Board 4.1.1 can Video Output Test by Encoder chip(CX25874)

7.1. Command

- [run 6]
- < Image file load to memroy>
- [init 0] (or 1, 2)

7.2. Video Output Test

- You must Development Board 4.1.1 Power On after connection D-SUB(or Video) cable.
- Execute [run 6] to start "DISPLAY test" mode.
- Load image file(800x600) into memory which will be displayed onto RGB Monitor(or TV).
 - a. Press Shift+F5 to stop the debugger.
 - b. Select "Load Memory From File" from the "File" menu in the debugger window.
 - c. Open "car_800x600_565.rgbbin" file in the supplied CD with the kit. (File location is DTK4 → SOFTWARE → PGM → BD_TEST → DIAGNOSTIC → IMAGE FILES.)
 - d. Put in address value in the pop up window to define the location of loaded image file.
(Example: 0x1000000)
 - e. The status bar in the bottom AXD window shows the memory loading status.
 - f. Wait until loading is completed. (Status bar will be disappeared when loading has completed)
- (After Project start again) Execute [init 0], and then you will show the image in to RGB Monitor(or TV)
- TV output and YPbPr output is same method too.

```

Input your touch > run 6
Start [TV/ARGB DISPLAY Test]

=====
< DISPLAY Test >
All parameter is hexadecimal!!!
=====
INIT [NUM]      : Initialize Display
  0              : ARGB 800x600Ix16x60Hz - cx25874
  1              : TV 720x480Ix16x30Hz - cx25874
  2              : YPbPr 720x480Px16x30Hz - cx25874
RB [ID] [ADDR]  : Read I2C Byte
WB [ID] [ADDR] [DATA] : Write I2C Byte
=====
[H] Help
[Q] Return
=====
DISPLAYTEST> init 0
CX25874 ID : 6, Rev : 3
SetDisplayMode : 5 mode - 0x01000000
DISPLAYTEST>

```

<Fig 18. A-RGB Test>

8. Camera Test

- Development Board 4.1.1 has CMOS Camera.
- Camera has 1/4", 1.3Mpixel, CMOS-Camera.
- If you want detail information, then Refer to Datasheet. (OV9640)

8.1. Command

- [run 7]
- [init 1]
- [a] (You will show the All register list of Camera)
- [rb 11] : Read byte register
- address : 0x11
- [wb 11 f0] : Write byte to register.
- address : 0x11, data : f0

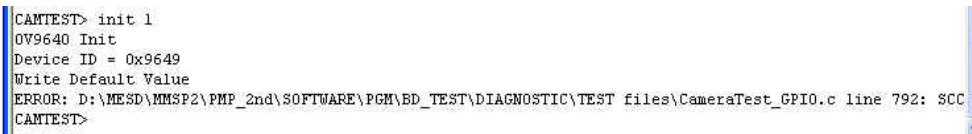
8.2. Camera Test

- Execute [run 7] to start "Camera test" mode.
- Execute [Init 1] to start camera initialize. (You will show the Preview image into LCD)
- Execute [a] to show the Register of Camera.
- You can change a register or read register. (RB and WB command)

```
Input your touch > run 7
Start [Camera Test]

=====
< CAMERA Test >
All parameter is hexadecimal!!!
=====
INIT [NUM] : Initialize Camera
0 (BAYER): OV9640 - ID : 0x60
1 (656P) : OV9640 - ID : 0x60
2 (656L) : OV9640 - ID : 0x60
RB [ADDR] : Read SCCB Byte
WB [ADDR] [DATA] : Write SCCB Byte
A : REG ALL
=====
[H] Help
[Q] Return
=====
CAMTEST>
```

<Fig 19. Camera Test>



```
CAMTEST> init 1
OV9640 Init
Device ID = 0x9649
Write Default Value
ERROR: D:\MESD\MMSP2\PM2_2nd\SOFTWARE\PGH\BD_TEST\DIAGNOSTIC\TEST files\CameraTest_GPIO.c line 792: SCC
CAMTEST>
```

<Fig 20. Init Command>

9. GPIO Interrupt Test

9.1. Command

- [Run 8]
- Press SW7, SW6, SW5, SW4
- [q] : exit

9.2. GPIO Interrupt Test

- Execute [run 8] to start “GPIO Interrupt Test” mode.
- Push button Switch Define
 - SW4 : Exit
 - SW5 : Falling Edge Interrupt
 - SW6 : Rising Edge Interrupt
 - SW7 : Low Level Interrupt

```

Input your touch > run 8
Start [GPIO Interrupt Test]

=====
< Switch(GPIO Interrupt) Test >
=====
SW1, SW2, SW3, SW4, SW5
=====
[SW1]      : Low Level Trigger
[SW2]      : Rising Edge Trigger
[SW3]      : Falling Edge Trigger
[SW4]      : Exit
[SW5]      : No Operation
=====
Preass Any Switch key
=====
GPIOC[11] Low Level Interrupt 0.K
GPIOC[11] Low Level Interrupt 0.K
GPIOC[12] Rising Edge Interrupt 0.K
GPIOC[12] Rising Edge Interrupt 0.K
GPIOC[13] Falling Edge Interrupt 0.K
GPIOC[13] Falling Edge Interrupt 0.K
GPIOC[13] Falling Edge Interrupt 0.K
GPIO Interrupt Test 0.K

If you want exit program, then Push the 'q' button.
: q

GPIO Interrupt Test Complete!!!

End [GPIO Interrupt Test]

```

<Fig 21. GPIO Interrupt Test>

10. SD Card Test

- Only Support SD Card Test. MMC Test does not support yet.
- Do not support Plug and Play at diagnostic. (You must insert a SD card before starting Test)
- Sub Board has 2 Slide Switch. It is choice Memory Stick and SD Card.

10.1. Command

- (after insert SD Card) [run 9]
- auto test

10.2. SD Card Test

- Previously insert SD Card.
- Execute [run 9] to start “SD Test” mode.
- Block Write/Read

```

Input your touch > run 9
Start [SD/MMC Test]

  * Card Initialize
In idle
CMD55 result = 1
CMD55 result = 1
CMD55 result = 1
In SD ready
End id
In stand-by
****4bit bus****
--> Done

  * Block Write Test
--> Done

  * Block Read Test
--> Done

--> O.K.

End [SD/MMC Test]
Input your touch >

```

<Fig 22. SD Card Test>

11. Ethernet Test

11.1. Command

- [run 10]
- [rd]
- Auto Test

11.2. Ethernet Test

- Execute [run 10] to start “Ethernet Test” mode.
- [RD] : PacketPage Pointer Register access to Write/Read
- Printed Result
- It is only checked communication between MMSP2 to Ethernet chip

```

Input your touch > run 10
Start [Ethernet Test]

=====
< Ethernet Test >
=====
Base Address : 0x84000300
=====
RD
A : 00 --> C : 630e
A : 20 --> C : 0300
A : 22 --> C : xxx4
A : 24 --> C : 3, 7, F
=====
all options are hexadecimal value
[H] Help
[Q] Return
=====
Ethernet Register > rd
----- Default Value -----
Register 0x8400030a = [3000 h]
Register 0x8400030c = [630e h]
----- After change -----
Write 0x00
Register 0x8400030a = [3000 h]
Register 0x8400030c = [630e h]

Write 0x20
Register 0x8400030a = [3020 h]
Register 0x8400030c = [0300 h]

Write 0x22
Register 0x8400030a = [3022 h]
Register 0x8400030c = [5554 h]

Write 0x24
Register 0x8400030a = [3024 h]
Register 0x8400030c = [5553 h]
Ok
Ethernet Register >

```

<Fig 23. Ethernet Test>

12. CF Card Test

12.1. Command

- [run 11]
- [cf]
- Auto Test

12.2. CF Card Test

- Execute [run 11] to start "CF Test" mode.
- [cf] : Waiting until insert CF card
- Attribute Area Read/Write when inserted CF Card

```
CF > cf
<CF TEST START>
- Waiting CF Card Inserted...
* CF Card Detect SLOT #0 [0x3e4f]
* [SLOT #0] CF Card Detect
* [SLOT #0] Supply 3.3V
* [SLOT #0] Wait CF Ready signal --> Ready OK
***** slot 0 *****

Tuple ID [0015]
* Major version = for 2.01/2.1
* Minor version = for 2.0/2.01/2.1
* Manufacturer = LEXAR ATA FLASH CARD
* Product name = TWISTER
* Product info 1 = TTE01

Tuple [001a] * Config Base Addr = 0x200

End of chain
* CFATA[0] Initialize
--> CFATA[0] Manufacture : LEXAR ATA FLASH
--> CFATA[0] Heads : 16
--> CFATA[0] Tracks : 984
--> CFATA[0] Sector : 32

* CFATA[0] PIO Mode Test
--> CFATA[0] PIO Test : [0.0] Percent finished.
--> CFATA[0] PIO Test : [12.5] Percent finished.
--> CFATA[0] PIO Test : [25.0] Percent finished.
--> CFATA[0] PIO Test : [37.5] Percent finished.
--> CFATA[0] PIO Test : [50.0] Percent finished.
--> CFATA[0] PIO Test : [62.5] Percent finished.
--> CFATA[0] PIO Test : [75.0] Percent finished.
--> CFATA[0] PIO Test : [87.5] Percent finished.
--> CFATA[0] PIO Test : [100.0] Percent finished.
```

<Fig 24. CF Test>

- A CF Card Recommended
 - SanDisk 128MB, 256MB, 512MB, 1G
 - Transcend 128MB, 256MB, 512MB, 1G
 - Hagiwara 128MB, 256MB, 512MB
 - RiData 128MB, 256MB, 512MB
 - Lexar 128MB, 256MB, 512MB
 - Samsung 128MB
 - Kodak 32MB

13. Touch Test

13.1. Command

- [run 12]
- Click a Touch Panel
- Auto complete after repeat 20 times

13.2. Touch Test

- Execute [run 12] to start “Touch Test” mode.
- Touch panel click.
- Printed a value of X,Y position.
- Auto complete after repeat 20 times

```
Input your touch > run 12
Start [Touch Test]

* TOUCH Test
Click Touch Panel
Test End Condition : Count = 20

--> X Position[0000], Y Position[1023], 1
--> X Position[0896], Y Position[0452], 2
--> X Position[0832], Y Position[0429], 3
--> X Position[0697], Y Position[0373], 4
--> X Position[0550], Y Position[0360], 5
--> X Position[0411], Y Position[0399], 6
--> X Position[0335], Y Position[0487], 7
--> X Position[0312], Y Position[0596], 8
--> X Position[0352], Y Position[0719], 9
--> X Position[0446], Y Position[0810], 10
--> X Position[0549], Y Position[0840], 11
--> X Position[0628], Y Position[0823], 12
--> X Position[0668], Y Position[0720], 13
--> X Position[0687], Y Position[0578], 14
--> X Position[0705], Y Position[0455], 15
--> X Position[0759], Y Position[0372], 16
--> X Position[0825], Y Position[0387], 17
--> X Position[0893], Y Position[0522], 18
--> X Position[0904], Y Position[0677], 19

End [Touch Test]
```

<Fig 25. Touch Test>

14. USB Device 1.1 Test

14.1. Command

- [run 13]
- [usb]
- Connect a Cable

14.2. USB Device 1.1 Test

- Execute [run 13] to start “USB Test” mode.
- [usb] : Waiting a connection
- Connect a Cable
- If the test was successful, then It will be Auto complete.
- But, If test was fail, then it will be waiting a connection. (This time, You should force close)

```

Input your touch > run 13
Start [USB Device Test]

-----
< USB 1.1 Test >
-----
USB : USB 1.1 Connect Test
-----
? or H                               : Help
Q                                     : Quit

USB Device 1.1 Test> usb

USB Device 1.1 Test...
Waiting a connection...
USB host is connected.

USB Device 1.1 Test> q

End [USB Device Test]

```

<Fig 26. USB Test>